Q1/ Define Artificial Intelligence

Q2/ What main capabilities does AI attempt to support

Q3/ List the major areas of AI

Q4/ List important applications of AI

Q5/ What are the main benefits of AI

Q6/ What are the risks and disadvantages of AI

Q7/ Compare human intelligence with artificial intelligence

Q8/ Define Machine Learning

Q9/ Compare traditional programming with Machine Learning

Q10/ What are the main steps of a Machine Learning project

Q11/ Define feature target and model

Q12/ Compare training validation and test data

Q13/ What are the main types of Machine Learning

Q14/ Compare classification with regression

Q15/ Define clustering

Q16/ Define reinforcement learning

Q17/ What metrics are commonly used for classification

Q18/ Why can accuracy be misleading

Q19/ Compare Machine Learning with Deep Learning

Q20/ How does data quality affect a model

Q21/ Name the components of a biological neuron and their functions

Q22/ Map biological neuron components to an artificial neuron

Q23/ Define an artificial neuron

Q24/ Write the artificial neuron equation

Q25/ What are the functions of weights and bias

Q26/ What is an activation function and why is it needed

Q27/ Compare the step and sign activation functions

Q28/ Define the linear activation function and its use

Q29/ Define Sigmoid and list its advantages and disadvantages

Q30/ Define tanh and compare it with Sigmoid

Q31/ Define ReLU and list its advantages and disadvantages

Q32/ List the key properties of neural networks

Q33/ List the advantages of neural networks

Q34/ List the disadvantages of neural networks

Q35/ Compare underfitting and overfitting

Q36/ Define generalization

Q37/ Calculate a step-neuron output for x=[1,0,1] w=[0.6,-0.4,0.8] and b=-0.7

Q38/ Calculate Sigmoid at z=0 and approximately at z=2

Q39/ What are the main components of a neural network architecture

Q40/ Define a single-layer feedforward network

Q41/ Define a multilayer feedforward network

Q42/ Define a recurrent network

Q43/ Define a competitive network

Q44/ How can one neuron implement AND

Q45/ How can one neuron implement OR

Q46/ How can one neuron implement NOT

Q47/ Why can one linear neuron not implement XOR

Q48/ Define a learning rule

Q49/ Compare supervised unsupervised and reinforcement learning

Q50/ Define the Hebbian learning rule

Q51/ Define the Perceptron learning rule

Q52/ Define the Delta or LMS rule

Q53/ Define Winner Take All learning

Q54/ Compare batch stochastic mini-batch and online training

Q55/ Verify OR for x=(1,0) w=(1,1) and b=-1

Q56/ Design a step neuron for NAND